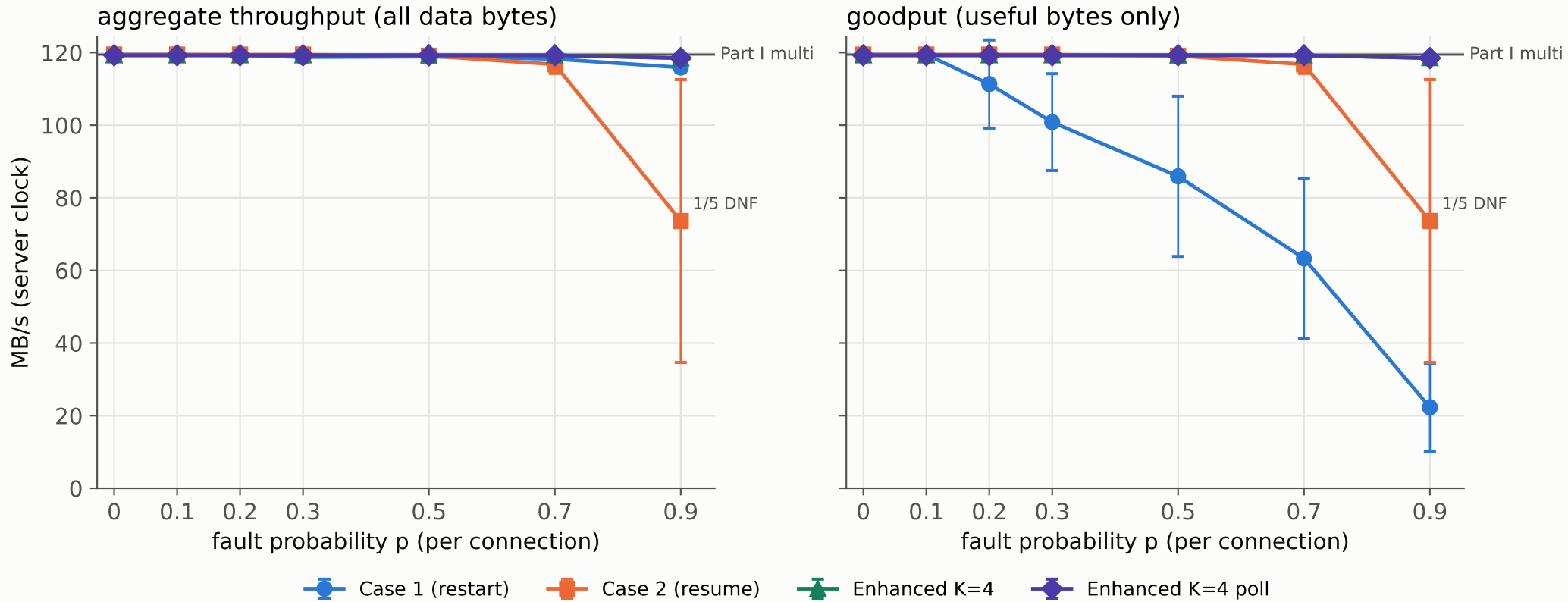Part II: all cases vs p